

Aim

To understand the UI layer in Android architecture by handling user interaction and updating UI using Activity and ViewModel.

Procedure

1. Open Android Studio.
2. Create a new project (Empty Activity, Kotlin).
3. Design UI in activity_main.xml using:
 - TextView (to display message)
 - Button (to trigger action)
4. Create a **ViewModel class** to manage data.
5. Open MainActivity.kt.
6. Connect UI components using findViewById.
7. Use ViewModel to fetch data.
8. Handle button click and update UI.
9. Run the app and observe output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tvTitle"
        android:text="UI Layer Demo"
        android:textSize="22sp"
        android:gravity="center"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/tvMessage"
        android:text="Message will appear here"
        android:textSize="18sp"
        android:paddingTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/btnLoad"
    android:text="Load Data"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

MainViewModel.kt

```
package com.example.uilayerdemo

import androidx.lifecycle.ViewModel

class MainViewModel : ViewModel() {

    fun getMessage(): String {
        return "Hello from ViewModel"
    }
}
```

MainActivity.kt

```
package com.example.uilayerdemo

import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.activity.viewModels
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private val viewModel: MainViewModel by viewModels()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        val btnLoad = findViewById<Button>(R.id.btnLoad)

        btnLoad.setOnClickListener {
            tvMessage.text = viewModel.getMessage()
        }
    }
}
```

Result

The application successfully:

- Handles user interaction through UI layer
- Uses ViewModel to provide data

- Displays dynamic message on button click



The image shows a web browser window with a dark title bar containing the text 'ex8' and standard window control buttons. The main content area has a light green background. At the top, the text 'MADL Lab Flutter' is displayed. Below it, a large heading reads 'Welcome to MADL Lab'. Underneath the heading is a rectangular input field with a thin border and the placeholder text 'Enter your name'. At the bottom of the visible area is a button labeled 'Submit'.

MADL Lab Flutter

Welcome to MADL Lab

Enter your name

TONY

Submit

Hello TONY, Welcome to MADL Lab!